

Chapter 10: Systems of Linear Differential Equations & Eigenvalues

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Course: Applied Ordinary Differential Equations (ENGR 213) | **Textbook:** Dennis G. Zill (7th Ed., Chapter 10)

1. Executive Summary & First-Principles Philosophy

Physical systems rarely consist of a single isolated mass or single electrical loop. Coupled mechanical springs, multi-compartment pharmacokinetic models, and interconnected electrical networks produce **systems of coupled differential equations**.

In matrix normal form:

$$\mathbf{X}'(t) = \mathbf{A}\mathbf{X}(t)$$

Solving a linear system reduces to finding the **Eigenvalues** λ and **Eigenvectors** $\mathbf{v}$ of the coefficient matrix $\mathbf{A}$, transforming calculus into linear algebra!

2. Core Mechanics & Mathematical Engine

A. Matrix Formulation

For a 2×2 system:

$$\begin{pmatrix} x_1' & x_2' \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} x_1 & x_2 \end{pmatrix} \iff \mathbf{X}' = \mathbf{A}\mathbf{X}$$

B. Eigenvalue Problem & Characteristic Equation

Substitute trial vector solution $\mathbf{X}(t) = \mathbf{v}e^{\lambda t}$:

$$\mathbf{A}\mathbf{v}e^{\lambda t} = \lambda\mathbf{v}e^{\lambda t} \implies (\mathbf{A} - \lambda\mathbf{I})\mathbf{v} = \mathbf{0}$$

For non-trivial solutions $\mathbf{v} \neq \mathbf{0}$:

$$\det(\mathbf{A} - \lambda\mathbf{I}) = 0$$

Three characteristic cases dictate the geometry of the phase plane:

1. Distinct Real Eigenvalues ($\lambda_1 \neq \lambda_2$):

$$\mathbf{X}(t) = c_1\mathbf{v}_1e^{\lambda_1 t} + c_2\mathbf{v}_2e^{\lambda_2 t}$$

- Both $\lambda > 0$: **Unstable Node (Source)**

- Both $\lambda < 0$: **Asymptotically Stable Node (Sink)**
 - Opposite signs $\lambda_1 > 0 > \lambda_2$: **Saddle Point (Always Unstable)**
2. **Complex Conjugate Eigenvalues** ($\lambda = \alpha \pm i\beta$): Eigenvector $\mathbf{v}_1 = \mathbf{a} + i\mathbf{b}$. Then two real solutions are:

$$\mathbf{X}_1 = e^{\alpha t} [\mathbf{a} \cos(\beta t) - \mathbf{b} \sin(\beta t)], \quad \mathbf{X}_2 = e^{\alpha t} [\mathbf{b} \cos(\beta t) + \mathbf{a} \sin(\beta t)]$$

- $\alpha = 0$: **Stable Center (Elliptical Orbits)**
 - $\alpha < 0$: **Stable Spiral Sink**
 - $\alpha > 0$: **Unstable Spiral Source**
3. **Repeated Real Eigenvalues** ($\lambda_1 = \lambda_2 = \lambda$):
- If only one independent eigenvector exists, find generalized eigenvector $\mathbf{u}$ solving $(\mathbf{A} - \lambda\mathbf{I})\mathbf{u} = \mathbf{v}$:

$$\mathbf{X}_2(t) = (\mathbf{v}t + \mathbf{u})e^{\lambda t}$$

3. Curriculum-Grounded Visual Reference

corresponding entries in the vector on the left, we obtain the more familiar statement

$$x = c_1 e^{-t} + 3c_2 e^{4t}, \quad y = -c_1 e^{-t} + 2c_2 e^{4t}.$$

As pointed out in Section 10.1, we can interpret these equations as parametric equations of a curve or **trajectory** in the xy -plane or **phase plane**. The three graphs shown in **FIGURE 10.2.1**, $x(t)$ in the tx -plane, $y(t)$ in the ty -plane, and the trajectory in the phase plane, correspond to the choice of constants $c_1 = c_2 = 1$ in the solution. A collection of trajectories in the phase plane as shown in **FIGURE 10.2.2** is said to be a **phase portrait** of the given linear system. What appears to be two red lines in Figure 10.2.2 are

Figure 10.1: Phase plane trajectories showing directional flow along eigenvector asymptotes — from Zill 7th Ed. Chapter 10 (Fig. 10.2.2).

4. Rapid Exam Traps & Red Flags

- **⚠ Trap 1: Finding the Generalized Eigenvector:** In defective repeated eigenvalue cases, the second solution is $(\mathbf{v}t + \mathbf{u})e^{\lambda t}$, **NOT** $\mathbf{v}te^{\lambda t}$. Forgetting $\mathbf{u}$ fails to satisfy $\mathbf{X}' = \mathbf{A}\mathbf{X}$.
- **⚠ Trap 2: Determinant Formula Sign:** For a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $\det(\mathbf{A} - \lambda\mathbf{I}) = \lambda^2 - \text{Tr}(\mathbf{A})\lambda + \det(\mathbf{A})$. Watch the signs of trace and determinant!